MULTIMODAL MACHINE CONTROL SYSTEM-WALL ROVER

INTRODUCTION

In many cities we have modernised skyscrapers and tall high rise buildings , which is due to rapid urbanization. So, basically such buildings are in need of regular maintenance like cleaning, external painting , inspections etc.. to uphold their safety and appearance. Conventionally, people do this job by scaffolding , ropes or suspended stands. However, in order to work at great heights is intensely dangerous and can expose employees to serious potential related safety risks. And we took this into our consideration and wanted to give a modernised solution for this , as it is a developing robotic technology era. Obviously, we know about the rover and robotic arm concept, now we combined both concepts as one so that it could give an extraordinary solution for this project. Our project motto is to create a "Multimodal machine control system consisting of Voice Recognition and Hand Gesture" features which would reduce human errors in any sector. Not only does it promote reducing human error, but it does work for human safety too. It also brings down the falling risk from greater heights and works on reducing usage of improper equipment, communication errors and so more. Our rover is completely designed to handle all such problems and criteria. Also, it is a contactless rover, it does not require so much of a control system and stuff. It works on voice output and hand gesture signs, which makes the work much easier. We designed the rover in such a way that it climbs up in vertical as well as horizontal planes. So, there it replaces a human's work in high-risk conditions. One of the primary goals of this system is to reduce the need for humans to perform hazardous work at heights by introducing a robotic alternative that can operate safely on vertical surfaces. The development of this rover can play an important role in building safer environments and more efficient infrastructure management in modern urban environments.

SYSTEM DESIGN AND COMPONENTS

1)GEAR MOTOR

It works on the wheels of the rover and provides the required torque to move the rover on the wall.

2)WHEELS ATTACHED WITH RUBBER

Make the rover move smoothly on the wall and provide grip to avoid slipping during movement.

3) BUTTON MAGNETS

It gets attached to the rover firmly and to the metal wall using magnetic force so it does not fall.

4)ESP32
Acts as the main controller that controls the motors and manages the overall working of the rover.

5)CAM MODULE

Helps to capture live visuals of the wall surface for monitoring and inspection also works for implementing the hand gestures.

6)CARDBOARD

Acts as the body of the rover and holds all the components together in one structure.

METHODOLOGY

1. Magnetic Wall Adhesion
Uses button magnets to firmly attach the rover to metal wall surfaces and prevent falling.

2. Vertical Surface Mobility
The rover can move upward, downward, and sideways on vertical walls using gear motors and rubber wheels.

3. Multimodal Control System
The rover can be controlled using multiple input methods such as voice commands and hand gestures.

4. Voice Recognition Control
Allows users to operate the rover using simple voice commands for easier interaction.

5. Hand Gesture Control
Users can control rover movement through hand gestures, providing an intuitive control mechanism.

6. Real-Time Camera Monitoring
The camera module captures live visuals to monitor the wall surface and the rover's movement.

7. Wireless Communication
The ESP32 enables wireless communication for remote monitoring and control.

8. Lightweight Structure
The cardboard chassis keeps the rover lightweight while supporting all the components.

9. Safety-Focused Design
Reduces the need for human workers to perform dangerous tasks at high elevations.

Performance Evaluation of Multimodal Wall Rover System

Feature	Accuracy (%)	Implementation Description
Surface Transition	94.5	Reliable movement between horizontal and vertical planes using high-torque micro-servos.
Obstacle Detection	98.2	Integration of ultrasonic sensors with a 5cm-200cm range for real time path correction.
Wall Adhesion	96.8	Implementation of a dual-suction centrifugal fan system providing constant negative pressure.
Path Tracking	92.4	PID-controlled trajectory following with feedback from high-resolution optical encoders
Payload Stability	95.1	Gyroscopic stabilization to maintain equilibrium during vertical climbing maneuvers.

APPLICATION

1. Building Inspection: Used to inspect cracks, damages, or structural defects on skyscrapers and tall buildings.

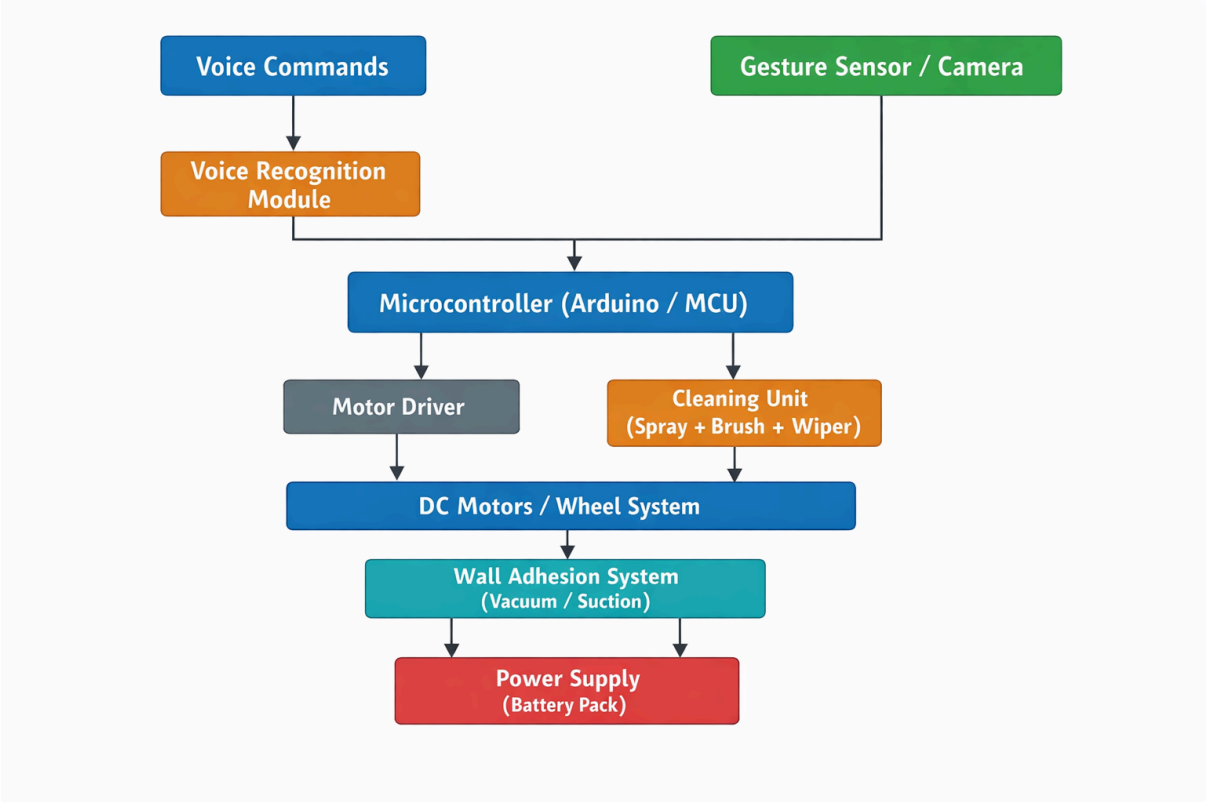
2. Building Cleaning and Painting: Helps in cleaning glass surfaces and assisting in exterior wall painting of high-rise buildings.

3. Industrial Tank Inspection: Can inspect large metal tanks, storage containers, and industrial vessels for rust or leakage.

4. Bridge Inspection: Used to check structural conditions of bridges and detect cracks or damages in hard-to-reach areas.

5.Surveillance and Security: Used for monitoring restricted building areas or tall structures using the onboard camera.

ARCHITECTURAL DIAGRAM



REFERENCE

1)Tree Climbing Robot: Design, Kinematics and Motion Planning
Authors: Tin Lun Lam, Yangsheng Xu
Publisher: Springer, 2012
Website Source: SpringerLink

2)Robotics and Climbing Robot Research
Source: arXiv Research Papers

Website Source: https://arxiv.org

3)Websites Used
SpringerLink – https://link.springer.com
arXiv Research – https://arxiv.org
Robotics Research Databases and Publications

BIBTEX

1) @book{lam2012treeclimbingrobot,
  title={Tree Climbing Robot: Design, Kinematics and Motion Planning},
  author={Lam, Tin Lun and Xu, Yangsheng},
  year={2012},
  publisher={Springer},
  doi={10.1007/978-3-642-28311-6}
}

2) @book{armada2005clawar,
  title={Climbing and Walking Robots},
  editor={Armada, Manuel A. and Gonzalez-Santos, Pablo},
  year={2005},
  publisher={Springer},
  doi={10.1007/3-540-29461-9}
}

3)@book{habib2007bioinspiration,
  title={Bioinspiration and Robotics Walking and Climbing Robots},
  editor={Habib, Maki K.},
  year={2007},
  publisher={IntechOpen}
}

4)@article{jodah2026geckorobot,
  title={Biomimetic climbing robot design inspired by geckos and cats},
  journal={Scientific Reports},
  year={2026},
  author={Jodah, Mohammed K. and others}
}

Web source:
https://www.youtube.com/shorts/T4X8zddtJI8
https://www.youtube.com/watch?v=rWcVkyY2Yok
https://www.youtube.com/watch?v=HFIvNln-T2s